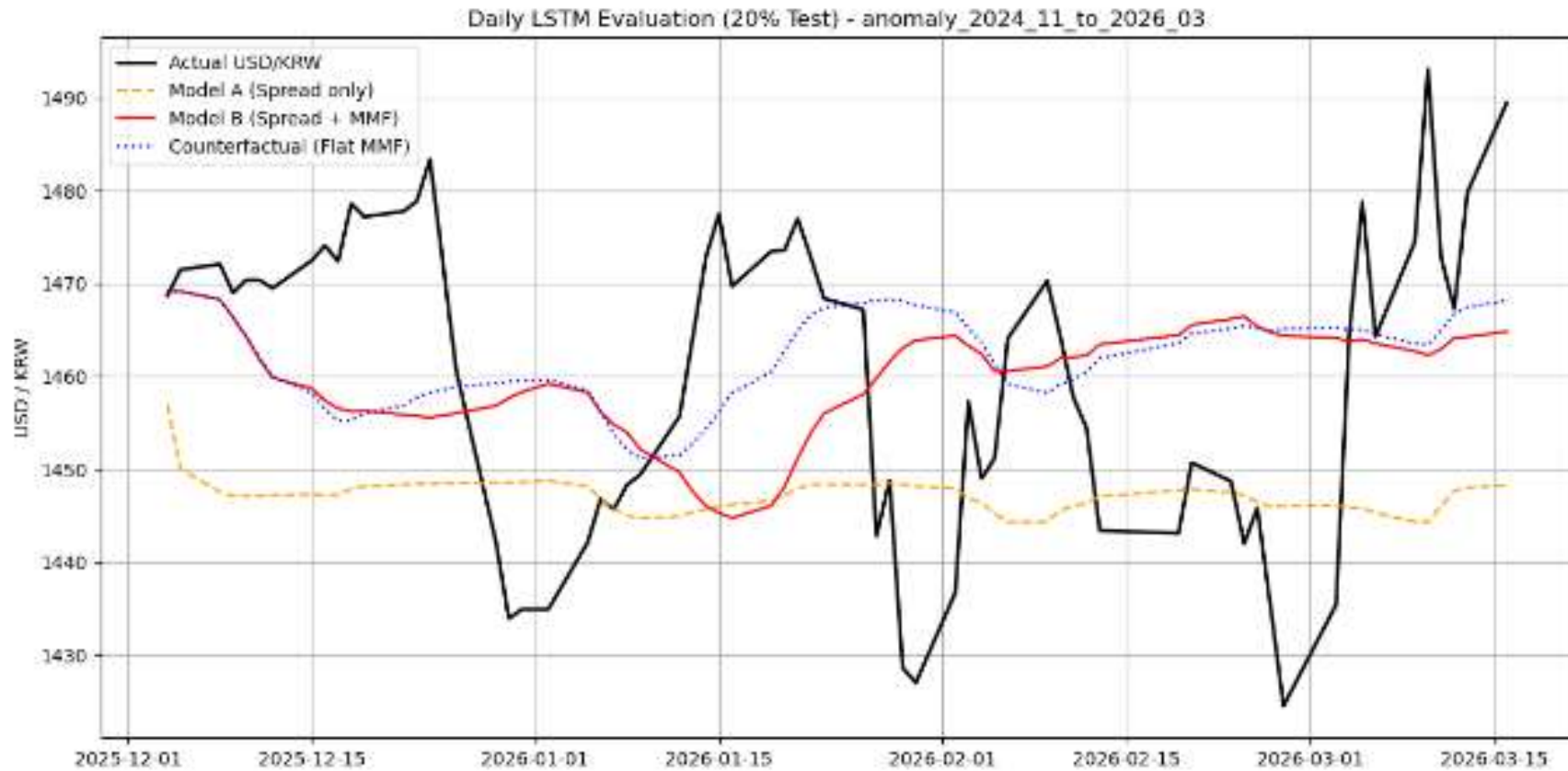


핀셋 2차 프로젝트

김예정 김한별 이지은

전주차 까지의 진행

환율 이상구간에서 M2와 강한 연관성 -> MMF 기반의 LSTM 모델
구축



문제점: 데이터의 부족 → 부족한 설명력

2024년 11월 ~ 2025년 12월 이상구간의 한계



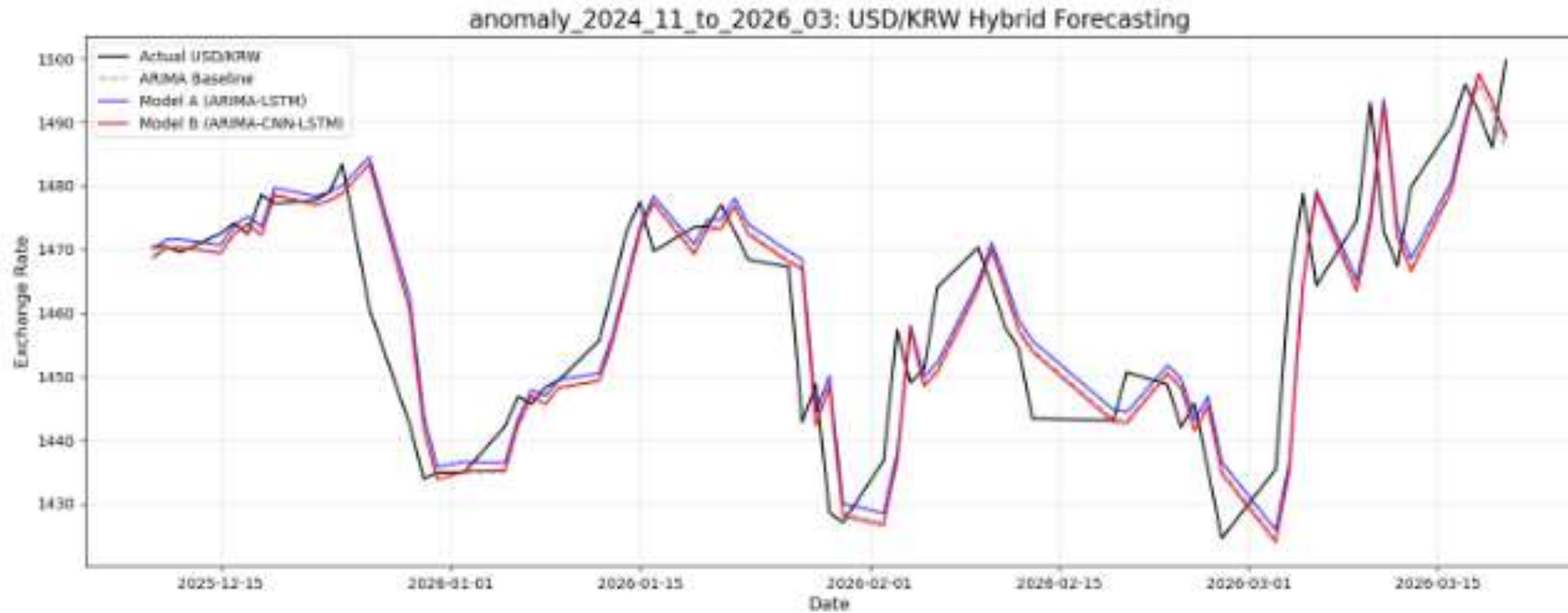
이상구간의 확장

~~고정된 기간~~ → **금리와 환율의 관계가 디커플링되는 모든 구간**



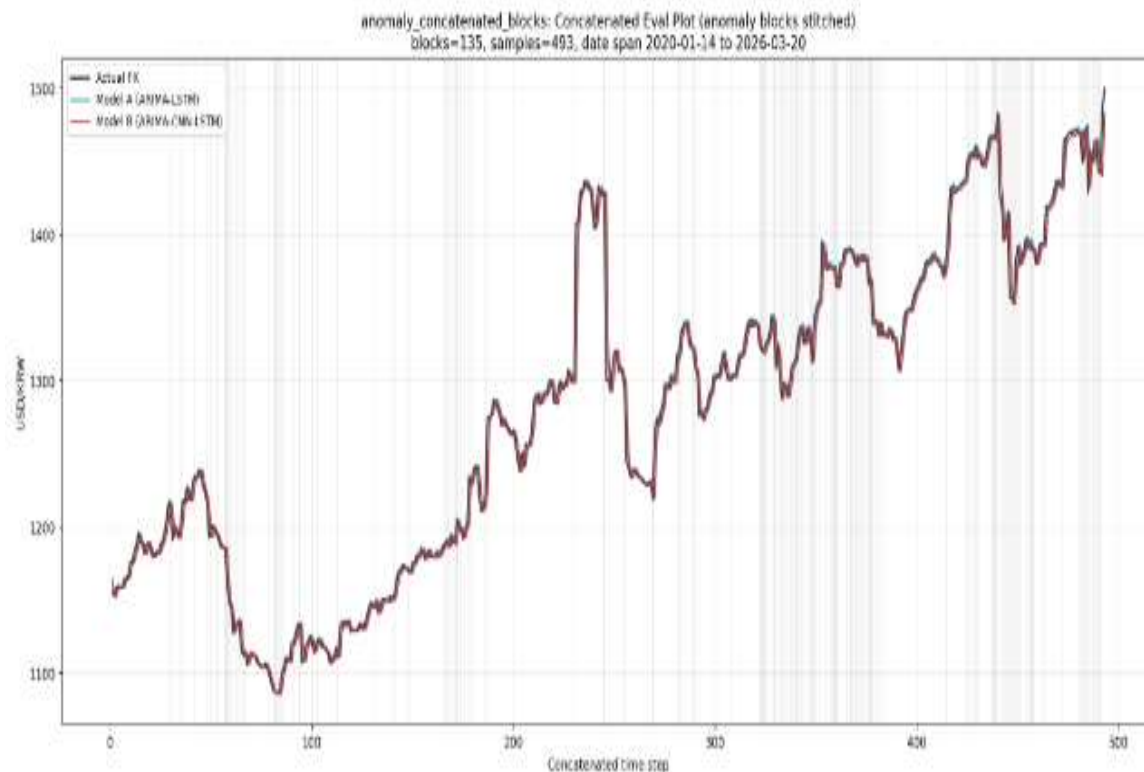
모델의 고도화

ARIMA - LSTM / ARIMA - CNN - LSTM



검증집합 일부에서의 성능

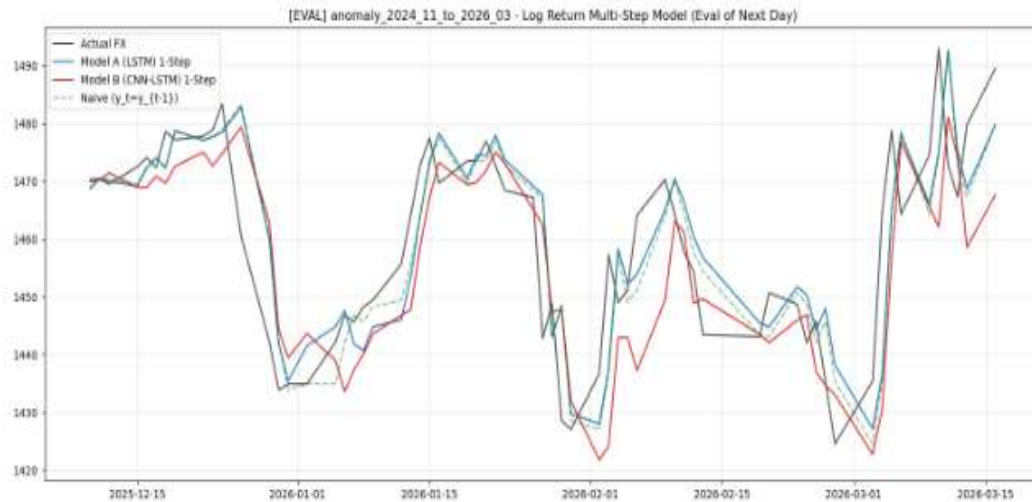
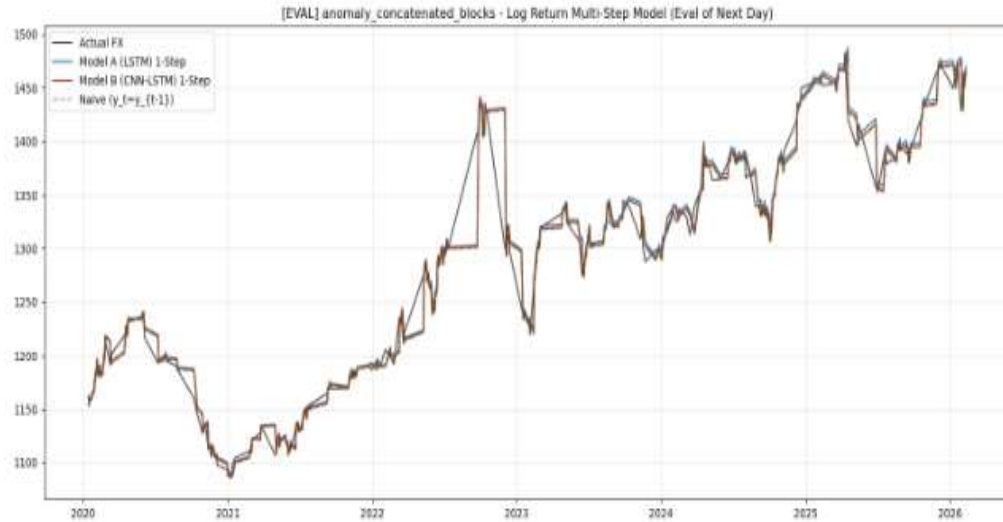
과적합 문제



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"period": "anomaly_2024_11_to_2026_03",  
"rows": 332,  
"train_rows": 265,  
"test_rows": 67,  
"rmse_base_arima": 9.934984753838158,  
"rmse_model_a": 9.966280451449975,  
"rmse_model_b": 9.954627744927487,  
"mae_model_a": 7.45093439176139,  
"mae_model_b": 7.490599572612962,  
"better_model": "B",
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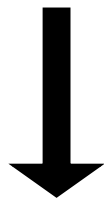
과적합 대응

로그 수익률 사용 + multi step 기반 예측



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"period": "anomaly_concatenated_blocks",  
"rows": 2467,  
"train_rows": 1974,  
"test_rows": 494,  
"rmse_base_arima": 12.77950813689083,  
"rmse_model_a": 13.080962577557806,  
"rmse_model_b": 12.847041218646211,  
"mae_model_a": 7.398441680734831,  
"mae_model_b": 7.061746502653729,  
"better_model": "B",
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“이상구간 환율 변동의 원인을 찾는다”



?

경상수지 CPI → 예측 + 영향 분석

```

1  TIME,CurrentAccount,GoodsAccount,ServicesAccount,PrimaryIncome,SecondaryIncome
2  1980-01-01,-600.4,-642.1,74.2,-67.7,26.2
3  1980-02-01,-548.1,-682.2,36.8,-130.3,27.6
4  1980-03-01,-676.0,-667.4,107.5,-155.0,34.1
5  1980-04-01,-617.1,-627.4,110.3,-153.0,33.0
6  1980-05-01,-541.2,-537.8,101.0,-133.1,38.7
7  1980-06-01,-450.5,-308.4,91.0,-295.5,43.0
8  1980-07-01,-610.4,-679.0,108.1,-104.3,33.7
9  1980-08-01,-516.0,-590.1,171.0,-124.0,32.4
10 1980-09-01,-430.5,-438.2,120.7,-160.3,30.3
11 1980-10-01,-487.5,-493.2,121.0,-164.4,40.5
12 1980-11-01,-501.3,-404.2,145.7,-101.0,38.2
13 1980-12-01,-810.3,-667.4,109.0,-140.7,43.2
14 1981-01-01,-115.1,-306.5,148.0,-118.6,31.1
15 1981-02-01,-507.7,-486.2,100.1,-157.1,29.5
16 1981-03-01,-470.0,-445.4,115.2,-206.2,37.5
17 1981-04-01,-612.0,-593.3,140.5,-201.0,35.0
18 1981-05-01,-710.6,-500.0,80.0,-205.2,32.3
19 1981-06-01,-350.2,-242.2,128.5,-173.5,40.0
20 1981-07-01,-663.1,-635.0,65.2,-172.1,70.6
21 1981-08-01,-401.4,-549.4,197.7,-103.5,41.8
22 1981-09-01,-303.7,-354.2,171.9,-250.4,30.0
23 1981-10-01,-219.5,-341.3,240.2,-182.8,64.4
24 1981-11-01,-622.7,-400.1,100.3,-354.5,43.6
25 1981-12-01,-1059.4,-1123.0,351.0,-330.5,32.0
26 1982-01-01,-344.0,-400.0,253.3,-141.4,42.8
27 1982-02-01,-100.2,-207.0,237.4,-110.0,40.0
28 1982-03-01,-270.1,-358.2,337.7,-306.8,49.2
29 1982-04-01,-415.3,-539.0,337.2,-200.0,47.1
30 1982-05-01,-387.4,-275.7,274.6,-233.5,47.2
31 1982-06-01,-614.0,-608.0,263.0,-321.7,42.1
32 1982-07-01,-202.5,-270.8,104.4,-167.4,61.3
33 1982-08-01,-520.4,-611.2,299.2,-144.9,62.5
34 1982-09-01,-562.1,-709.0,372.4,-268.3,42.8
35 1982-10-01,-780.0,-709.5,195.0,-312.8,40.1
36 1982-11-01,-786.1,-808.2,268.7,-287.7,41.1
37 1982-12-01,-710.0,-738.0,273.8,-327.4,54.8
38 1983-01-01,-476.6,-526.2,186.0,-179.7,43.3

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1  observation_date,CPI_KOR
2  2010-12-01,87.251
3  2011-01-01,88.288
4  2011-02-01,88.892
5  2011-03-01,89.238
6  2011-04-01,89.325
7  2011-05-01,89.497
8  2011-06-01,89.67
9  2011-07-01,90.102
10 2011-08-01,90.707
11 2011-09-01,90.62
12 2011-10-01,90.448
13 2011-11-01,90.534
14 2011-12-01,90.879
15 2012-01-01,91.242
16 2012-02-01,91.588
17 2012-03-01,91.614
18 2012-04-01,91.605
19 2012-05-01,91.769

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